DATA STRUCTURES LAB RECORD

Implementation of Kruskal's Algorithm in C

Source Code (KRUSKALS.C)

```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>

int i, j, k, a, b, u, v, n, ne = 1;
int min, mincost = 0, cost[9][9], parent[9];
int find(int);
int uni(int, int);

void main() {
    printf("\n\tImplementation of Kruskal's Algorithm\n");
    printf("\nEnter the no. of vertices:");
    scanf("%d", &n);
    printf("\nEnter the cost adjacency matrix:\n");
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            scanf("%d", &cost[i][j]);
            if (cost[i][j] == 0)
                cost[i][j] = 999;
        }
    }
    printf("The edges of Minimum Cost Spanning Tree are\n");
    while (ne < n) { // edges < vertex - 1
        for (i = 1, min = 999; i <= n; i++) {
            for (j = 1; j <= n; j++) {
                if (cost[i][j] < min) {
                    min = cost[i][j];
                    a = u = i;
                    b = v = j;
                }
            }
        }
        u = find(u);
        v = find(v);
        if (uni(u, v)) {
            printf("%d edge (%d,%d) =%d\n", ne++, a, b, min);
            mincost += min;
        }
        cost[a][b] = cost[b][a] = 999;
    }
    printf("\n\tMinimum cost = %d\n", mincost);
    getch();
}

int find(int i) {
    while (parent[i])
        i = parent[i];
    return i;
}

int uni(int i, int j) {
    if (i != j) {
        parent[j] = i;
        return 1;
    }
```

Sample Output

```
Implementation of Kruskal's Algorithm

Enter the no. of vertices: 4

Enter the cost adjacency matrix:
0 10 6 5
10 0 0 15
6 0 0 4
5 15 4 0

The edges of Minimum Cost Spanning Tree are
1 edge (3,4) =4
2 edge (1,4) =5
3 edge (1,2) =10

        Minimum cost = 19
```